# 2025_HL60_KO_Intracellular_truncation_data

June 8, 2025

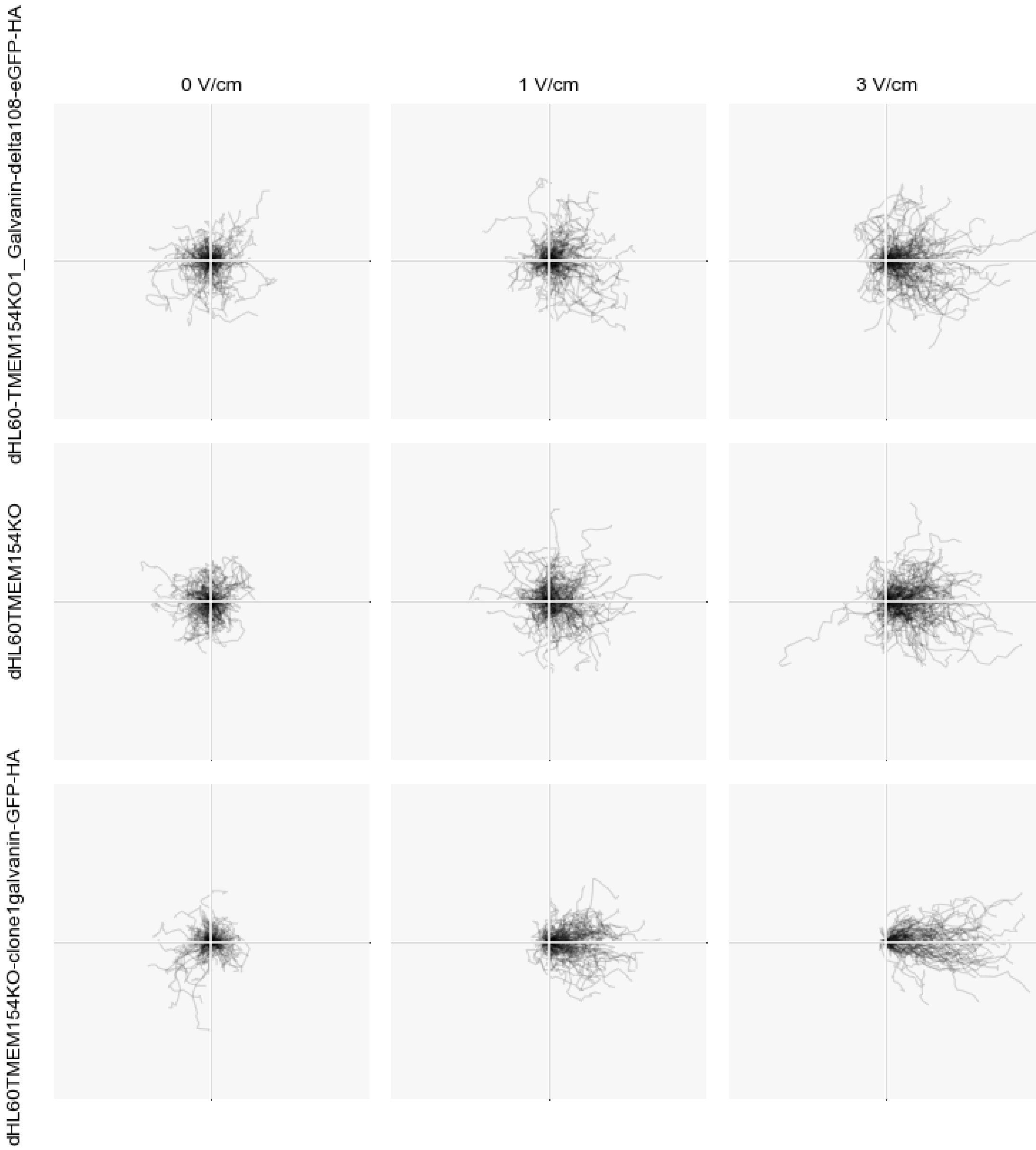
[2]:
```
array(['dHL60TMEM154KO', 'dHL60-TMEM154KO1_Galvanin-delta108-eGFP-HA',
       'dHL60TMEM154KO-clone1galvanin-GFP-HA'], dtype=object)
```

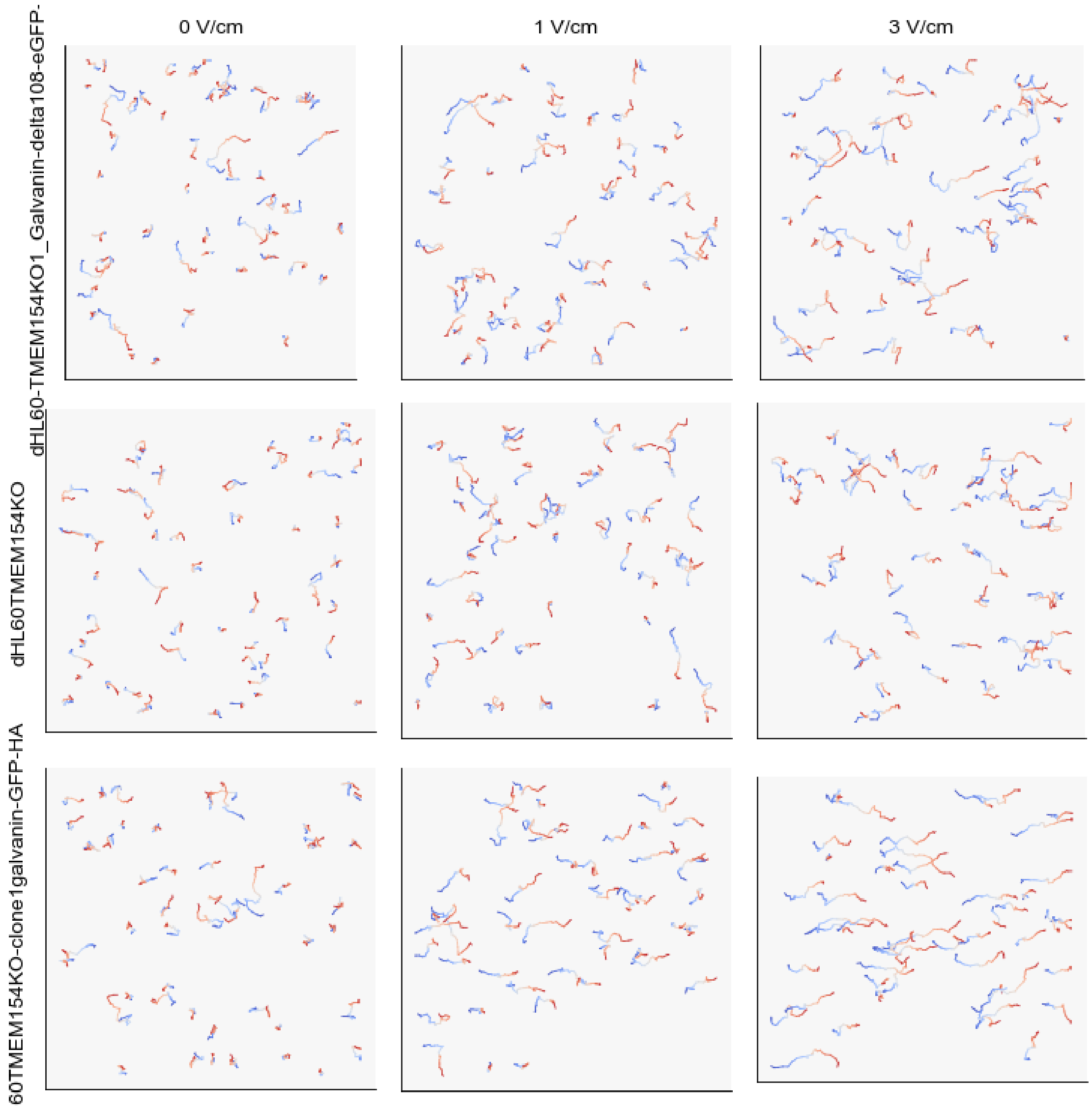
[3]:
```
                                     celltype  E_V_cm      date  trial user  \
0  dHL60-TMEM154KO1_Galvanin-delta108-eGFP-HA       0  20241119      0   AP
1  dHL60-TMEM154KO1_Galvanin-delta108-eGFP-HA       0  20241119      0   AP
2  dHL60-TMEM154KO1_Galvanin-delta108-eGFP-HA       0  20241119      0   AP
3  dHL60-TMEM154KO1_Galvanin-delta108-eGFP-HA       0  20241119      0   AP
4  dHL60-TMEM154KO1_Galvanin-delta108-eGFP-HA       0  20241119      0   AP

   cell  frame         x          y           z
0     5      0  0.000000  13.679891  180.000000
1     5      1  0.139378  14.248139  173.707215
2     5      2  5.756859  15.191881  170.192601
3     5      3  5.652926  14.673184  168.428756
4     5      4  0.868936  11.942501  166.037317
```

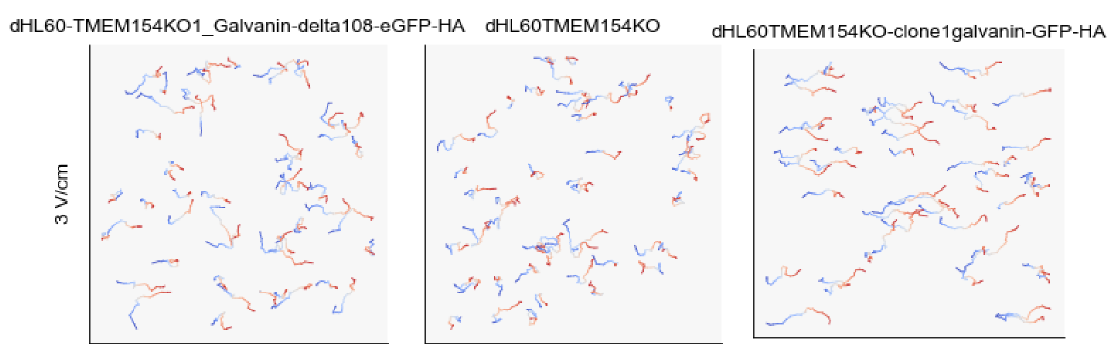
# 1 Plotting of HL-60 Galvanin constructs where we clip off the intracellular domain

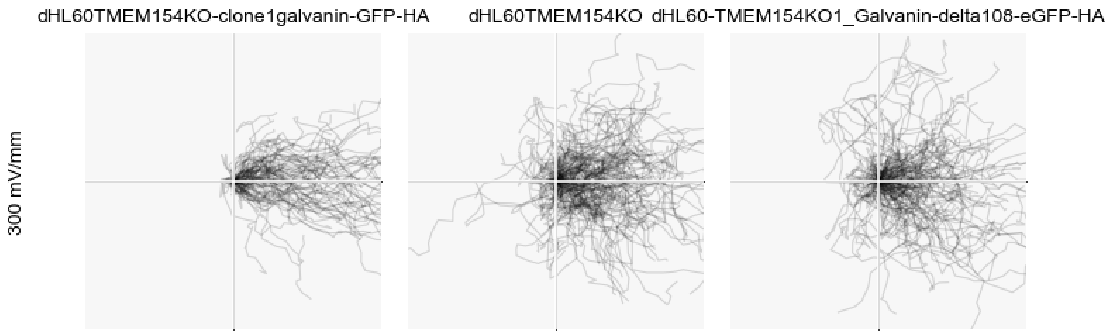
## 1.1 Tracks



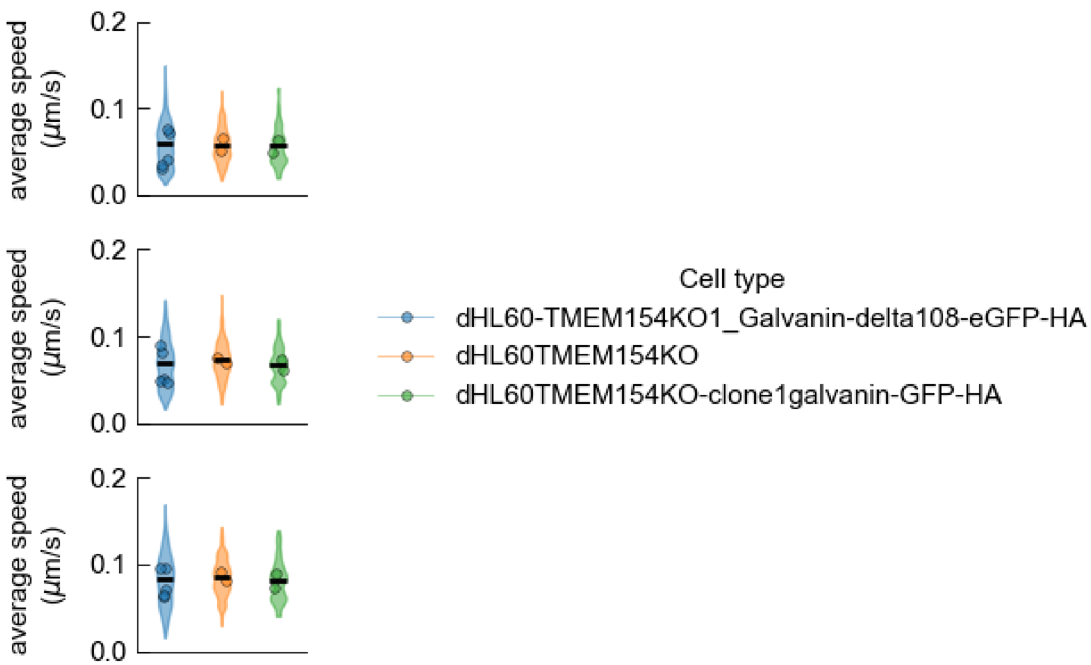


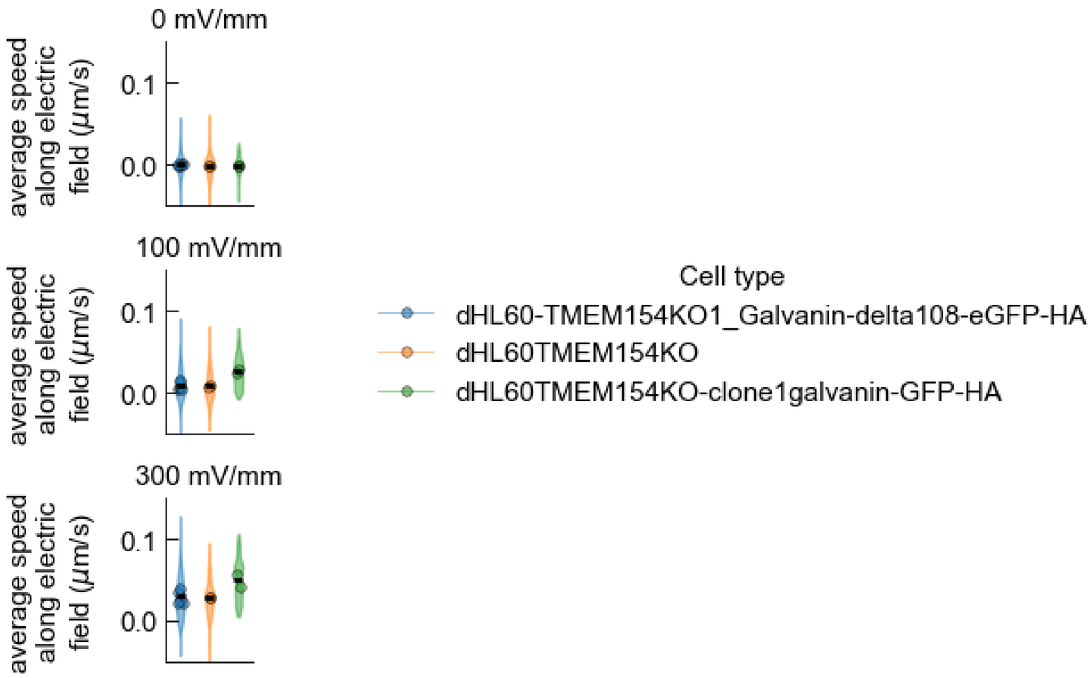

### 1.2 Here are only plots of 300 mV/mm condition only - probably more useful for main text figures.



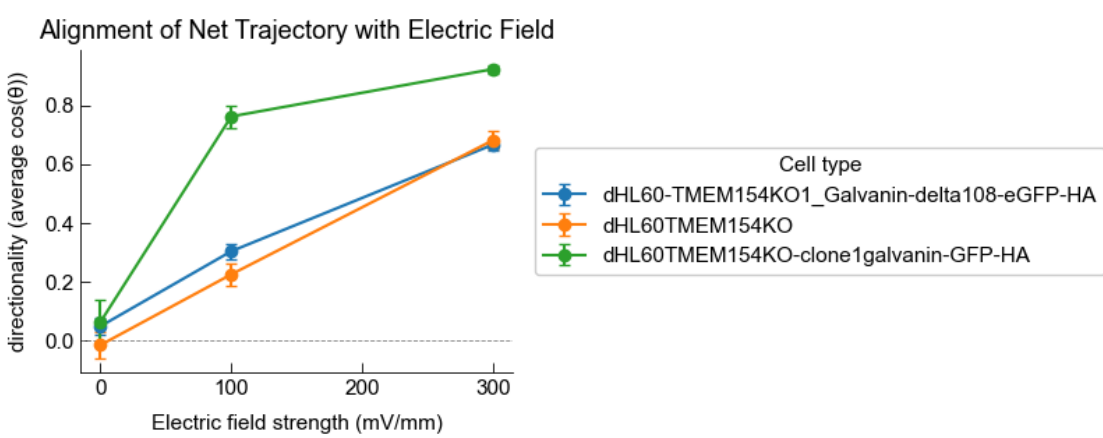



## 2 Speed related



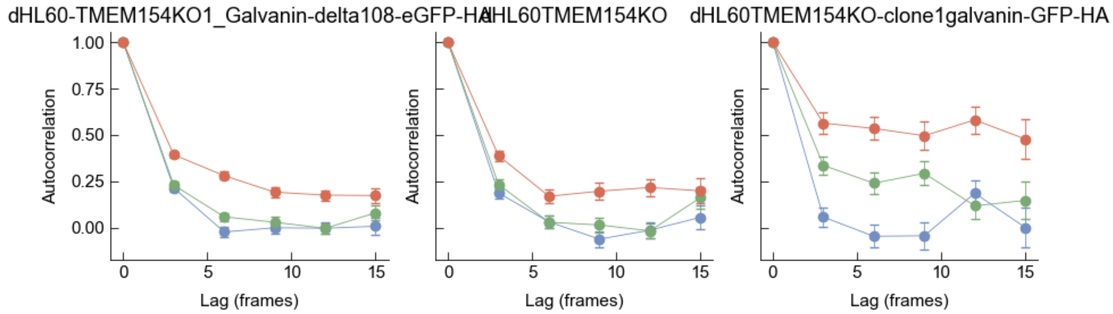


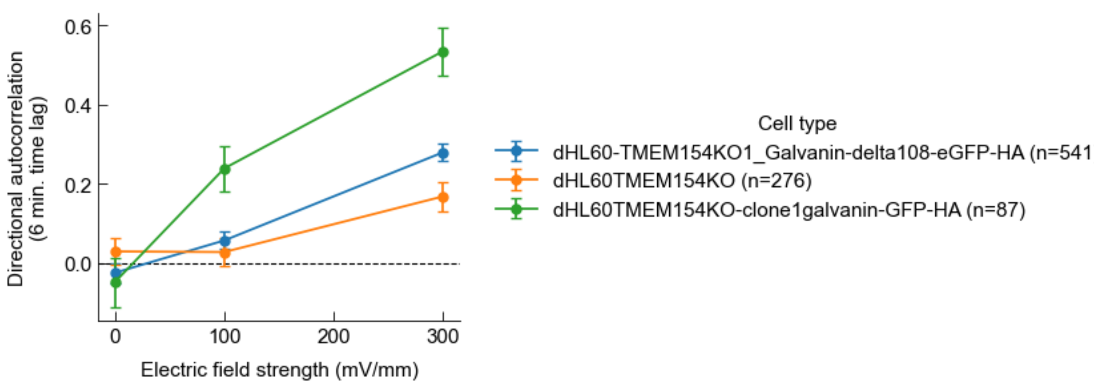

## 3 Directionality (average cosine theta)




## 4 Autocorrelation Analysis

This uses cosine theta between trajectory vector and electric field (the regular approach we've been taking).




## 4.1 Autocorrelation, but relative to 0 mV/mm

Here we plot relative to the 0 mV/mm condition to try to account for residual persistaence in the absence of an electric field. (i.e. subtraction of autocorrelation value from the 0 mV/mm condition).